Semiconductors: New Technologies and the Value Chain

Module 4: Integrated circuit design

Live Session

July 2026

Learning Outcomes

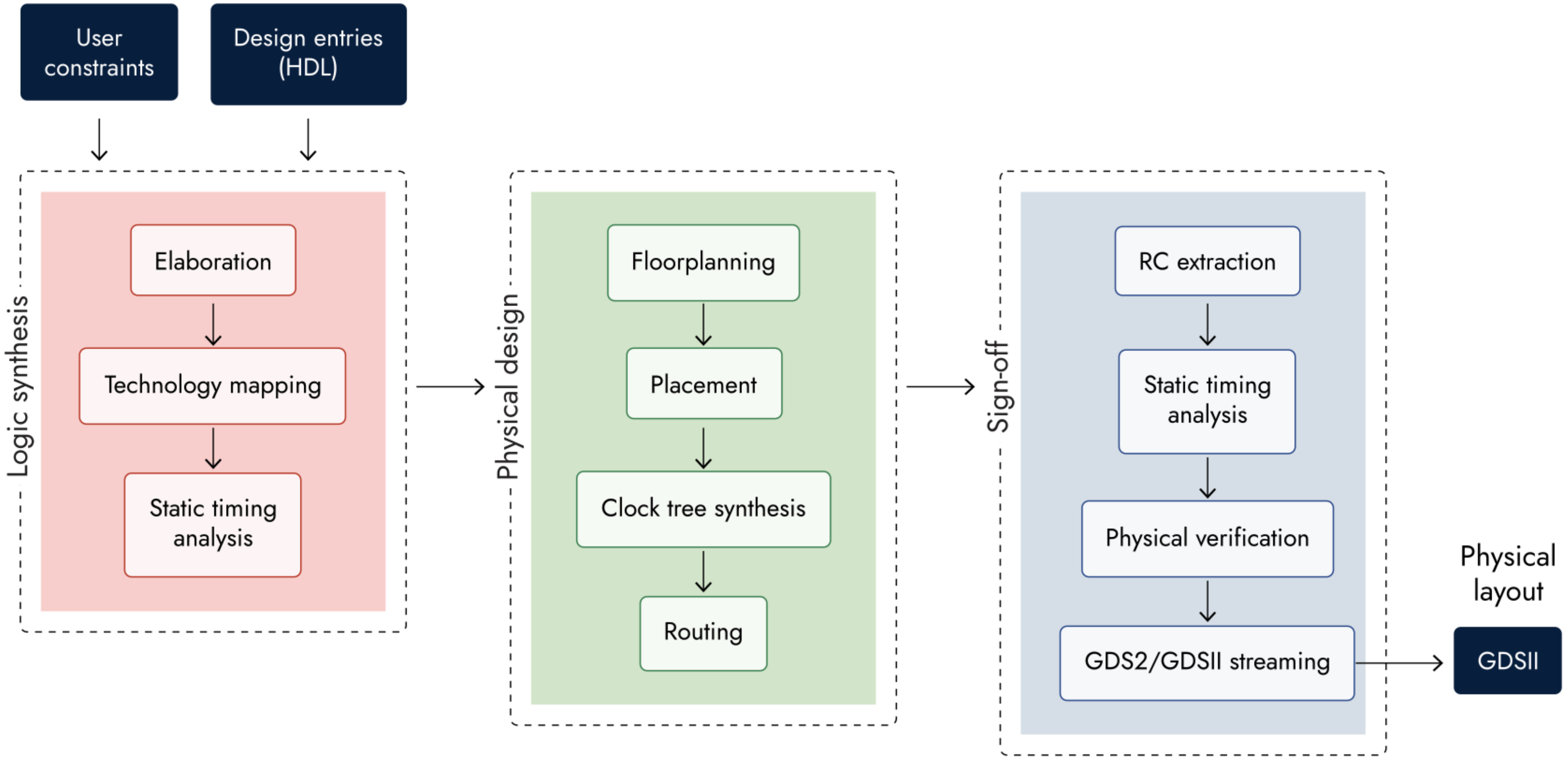
By the end of this module you will be able to:

- Acquire skills to use IC automation tools to generate a physical implementation from a design.
- Compare different business models around IC design and manufacturing.
- Evaluate the challenges in the semiconductor value chain holistically.

A warm-up question

What are the challenges in IC design?

- Design complexity – number of gates grow with Moore's law
- Business models – Fabless + Foundry or IDM
- Design for manufacturability and testability



ASIC design process

Energy and Power in IC Design

$$E = C_L V_{DD}^2 P_{0\to 1} + t_{sc} V_{DD} I_{peak} P_{0\to 1} + V_{DD} I_{stat}$$

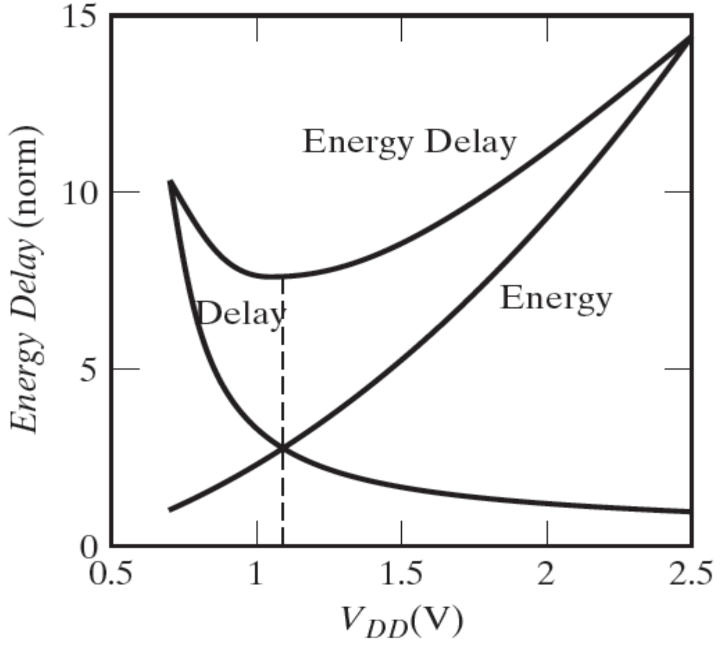
$$P = C_L V_{DD}^2 f_{0\to 1} + t_{sc} V_{DD} I_{peak} f_{0\to 1} + V_{DD} I_{stat}$$

Leakage power
~15-35% today

Switching power
60-80% today

Short-circuit power
<5-10% today

Trade-off between energy and delay



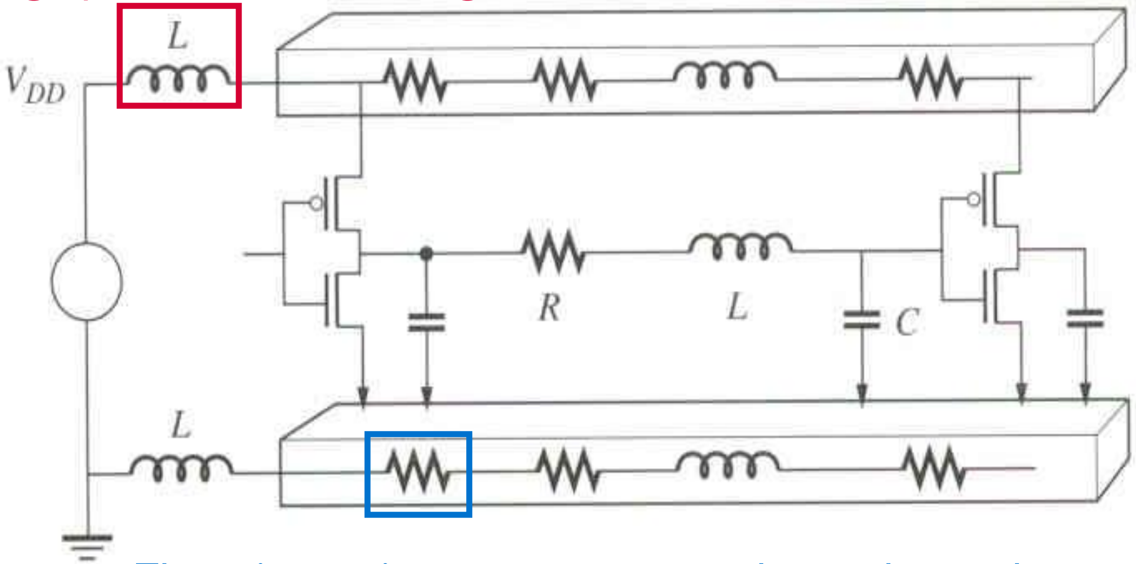
- Dynamic voltage and frequency scaling
  - lower V_{DD} and $f_{0\to 1}$ to save energy (e.g. Automotive TCU example)
- EDA keeps t_{sc} short by adding in buffers if required.
- I_{stat} is minimised by
  - 3D Transistors (FinFET & GAAFET/Nanosheet)
  - High-*k* dielectrics & Metal Gates (HKMG)
  - Power gating (design stage): cuts off supply voltage entirely to idle core blocks.

Power Distribution System

As current is acquired from the supply in a short period of time, the inductance due to the package pin results in voltage drop.

Wires exhibits parasitic *R*, *L*, *C* depending on their geometries.

The power supply pins on the IC package drive a huge network of wires to distribution power to the transistors.



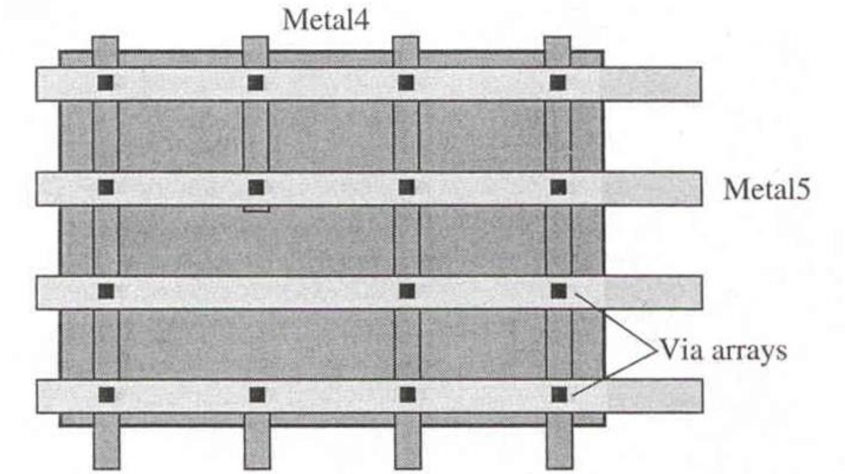
At any point of the supply, there would be a potential voltage drop ΔV

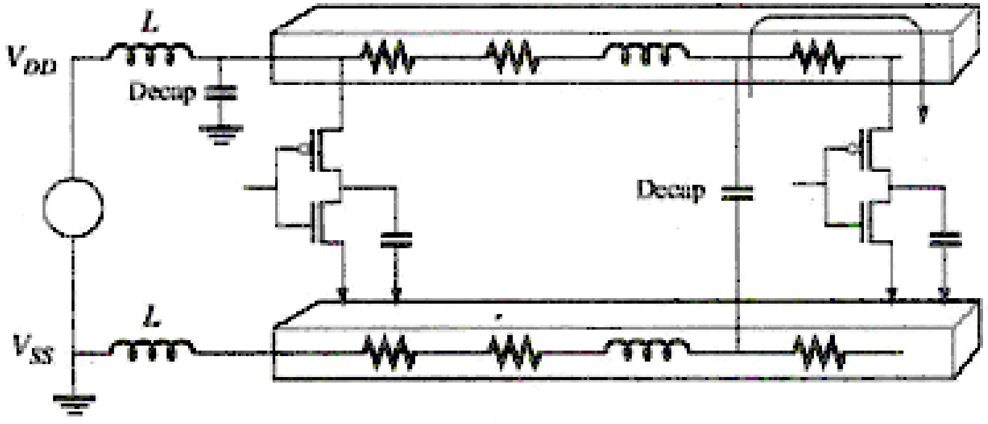
$$\Delta V = IR + L\frac{di}{dt}$$

The wire resistance creates voltage drops that increase as the current moves towards the inverters. It slows down the logic gates and compromises the noise margins.

Solutions to IR & Ldi/dt Drop

- Reduce *R* by keeping a short distance between supply pins and the circuit supply through a structured layout of the power distribution network.
- Decoupling capacitors (Decap): large capacitors that acts a reservoir of charges near the power pins and any drivers
- Reduce *L* by using different package technologies and keep interconnect short.





Breakout Group – Reimagining transistor structures

- Take turns to discuss your thoughts:
  "What can we do with chip defects?"

  It may be useful to consider from the upstream, midstream and downstream perspectives of semiconductor value chain
- Time: 15 minutes

Questions and Answers

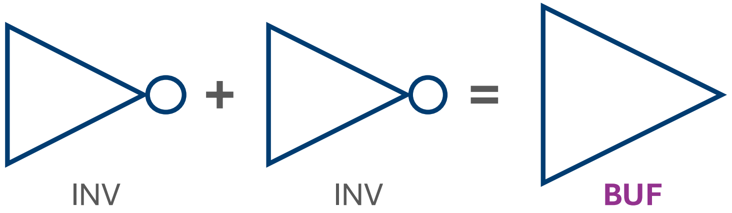
- Do you have any questions from today's discussion or the Canvas materials?

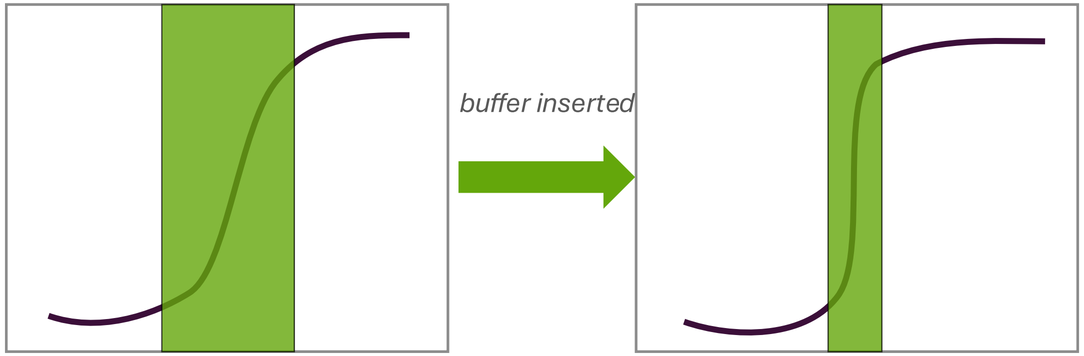
References

Chapters 13.3 and 13.4
CMOS VLSI Design: A Circuits and Systems Perspective (4th edition)
by Neil Weste and David Harris

Buffers: Reducing t_{sc}

Buffer = two inverters in series (schematic symbol)





Before buffer: slow edge

→ **long t_{sc}**

After buffer: sharp edge

→ **short t_{sc}**

Shaded band = both transistors briefly ON (the t_{sc} window)

- Buffer = a non-inverting logic cell (built from two inverters in series) that restores signal strength and sharpens edges.
- A signal travelling over a long, resistive and capacitive wire loses sharpness: slow RC charging.
- Then a slow edge keeps NMOS and PMOS both conducting for longer during the transition and this overlap is t_{sc}.
- Inverter is the most effective gate to produce output current due to its simplicity and speeds up the RC charging.
- EDA tools insert a buffer along the path to restore a fast, sharp edge before the next gate shrinking t_{sc} and short-circuit power.